

HARDWARE

Hardware is the first stop on the Tweak Express. A few right and quick fixes, and your system will surprise you



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Quickly press **[Delete]** as your computer starts up to enter your PC's BIOS. This is where you set your system for better performance—no matter what OS you're running.

Flashers

Before anything else, if your BIOS version is over a year or two old, chances are your motherboard manufacturer has released an updated version that will significantly increase performance and stability. If the flashing process is interrupted, your motherboard will be rendered useless: flashing is therefore not for the faint hearted. Get your system assembler to update your BIOS.

Beyond clock speeds

The first step to overclocking your CPU is to increase the Front Side Bus (FSB). Before you start, ensure you have a well-ventilated cabinet, with at least three fans—one for ventilation, one for the Switched Mode Power Supply (SMPS) and one for the CPU. Increase the FSB in the smallest of instalments—from, say, 133 MHz to 135 MHz. Now change your CPU ratio marginally, and keep testing system stability and performance. Of course, this only applies to Athlon CPUs.

Intel locks theirs to prevent overclocking accidents. Don't give in to the temptation of trying to

change voltage settings—this will significantly increase the amount of heat produced.

Basic BIOS tweaks

Here are some generic BIOS tips to increase performance. The gains are limited, but the tweaks aren't as risky as overclocking.

Firstly, set the First Boot Device to your C: drive and disable the Second, Third and Other boot devices. On the rare occasions when you need to boot from a floppy or CD, a deft move on [Delete], and changing the First Boot Device option to Floppy or CD will remedy the situation.

Change the 'CAS Latency Time' from 3 to 2—this reduces the amount of time your RAM waits before it starts a 'read' command.

Change the 'RAS-to-CAS Delay' and the 'RAS Pre-charge Time' to 2—this reduces the delay between signals and lets the memory refresh faster.

Make sure that the bootup virus checking option is disabled to increase boot times significantly.

Disable 'Boot up Floppy seek' so that the BIOS won't waste time looking for a floppy drive. Disable the Video BIOS Shadow.

If you have a graphics card with over 16 MB of memory, disable 'Video RAM Cacheable'.

Set the initial display card as either PCI or AGP, and not 'Auto', depending on the configuration of your graphics card.

Dirty hardware

Cleanliness is next to godliness, even in technology. Keep your cabinet clean and in a well-ventilated area. This is especially important for surfaces such as those of the CPU, GPU and power supply that heat up. As long as these are clean, the heat won't build up.

Use soft paint brushes to clean fans to prevent them from slowing down or locking up, which can cause hardware damage.

Cable chaos

Arrange and tie your cables with rubber bands or cotton threads to prevent entanglement and improve ventilation. Also, try and space hard drives as far away from each other as possible. Consider adding a fan for your hard drives if your cabinet is too small.

Separate IDEs please

A common assumption is that if the hard disk and CD-ROM drive are on the same cable, data transfer will be faster. This isn't true, since data transfer rates depend on Integrated Device Electronics (IDE) controllers, which run at the speed of the slower device. Since the optical drive is slower than the hard disk, connecting them to the same IDE port slows down performance.

Also, transfer speeds between drives on two different IDE channels is faster than that between devices on the same IDE channel.